Lab No: 03

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Roll No: 17

Task 1:	

Code:	void insertAtStart(string value)
	{
	node *temp =new node();
	temp->data = value;
	temp->next = head;
	head = temp;
	}
	
	void insertAtAnyPosition(string value,int pos)
	{
	node *temp =new node();
	temp->data = value;
	
	if (pos == 1)
	{
	temp->next = head;
	head = temp;
	return;
	}
	
	node *tem = head;
	for(int i=0; i<pos-2; i++)
	{
	tem = tem->next;
	}
	temp->next = tem->next;
	tem->next = temp;

	}
	
	void insertAtEnd(string value)
	{
	node *temp = new node();
	temp->data = value;
	temp->next = nullptr;
	
	if (head == nullptr)
	{
	head = temp;
	
	}else{
	
	node *tem = head;
	while(tem->next != nullptr)
	{
	tem = tem->next;
	}
	tem->next = temp;
	}
	}

Screenshots:	

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
1

Enter Value you want to insert at first : 0
```

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
3

Enter Value you want to insert at end : 2
```

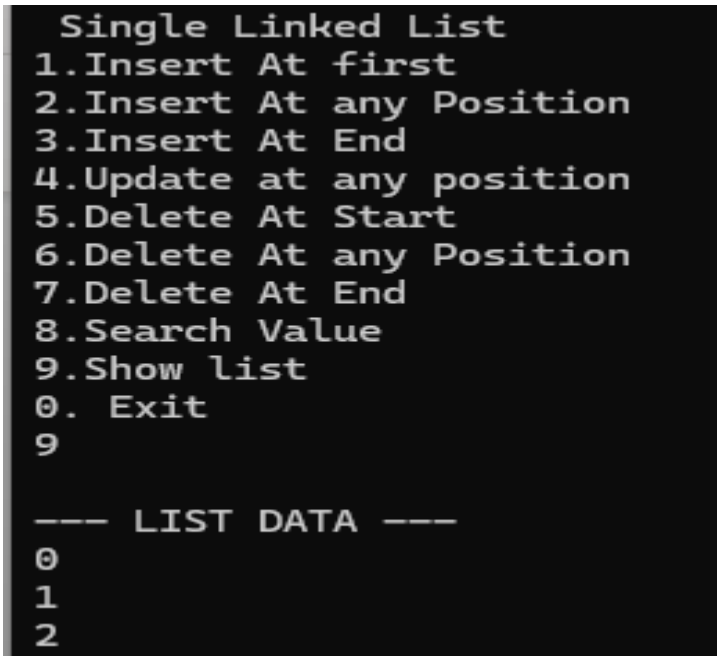
```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
2

Enter Value you want to insert at certain position : 1

Enter Position you want to insert : 2
```

Task 2:	

Code:	void showlist()
	{
	node *temp = head;
	if (temp == nullptr)
	{
	cout << "List is empty!" << endl;
	return;
	}
	while(temp != nullptr)
	{
	cout<<(temp->data)<<endl;
	temp = temp->next;
	}
	}

Screenshot:	

Task 3:	

Code:	void DeleteAtStart()
	{
	if(head == nullptr)
	{
	cout<<"List is Empty"<<endl;
	return;
	}
	else{

```
        node *Todelet = head;
        head = head->next;
        delete Todelet;
        return;
    }
}

void DeleteAtAnyPosition(int pos)
{
    if (pos == 1)
    {
      node *Delet = head;
     head = head->next;
     delete Delet;
     return;
    }

 else{
    node *prev = head;
    for(int i=0; i<pos-2; i++)
    {
      prev = prev->next;
    }
    node *Delet = prev->next;
    prev->next = Delet->next;
    delete Delet;
    return;
 }
}

void DeleteAtEnd()
{
    if(head==nullptr)
    {
          cout<<"List is Emty"<<endl;
          return;
    }

     if (head->next == nullptr)
    {
       delete head;
       head=nullptr;
       return;

    }else{
```

```
    node *prev = head;
    node *curr = head->next;
     while(curr->next != nullptr){
        prev = curr;
        curr = curr->next;
     }
         prev->next = nullptr;
        delete curr;
   }
}
```

Screenshot:

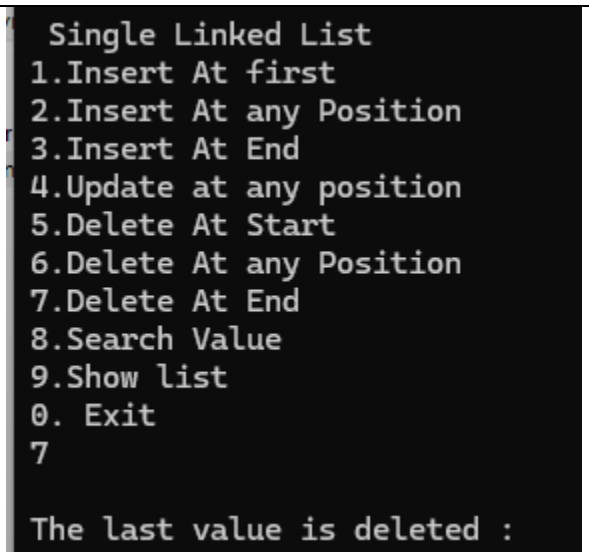
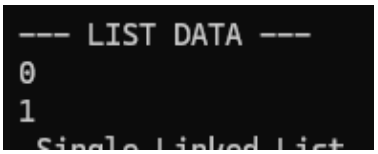
```
--- LIST DATA ---
hania
0
1
2
aira
```

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
5

The first value is deleted :
```

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
6

Enter Position which you want to delete : 3
```

Task 4:	

Code:	void SearchValue(string value)
	{
	node *temp = head;
	int pos = 1;
	
	while(temp!=nullptr)
	{
	if(temp->data == value)
	{
	cout<<value<< " Exist at Positon : "<< pos<<endl;
	return;
	}
	temp = temp->next;
	pos++;
	}
	cout<<value<<" does not exist. "<<endl;
	}

Screenshot:	

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
8

Enter Value you want to Search in the list : 2

2 does not exist.
```

```
 Single Linked List
1.Insert At first
2.Insert At any Position
3.Insert At End
4.Update at any position
5.Delete At Start
6.Delete At any Position
7.Delete At End
8.Search Value
9.Show list
0. Exit
8

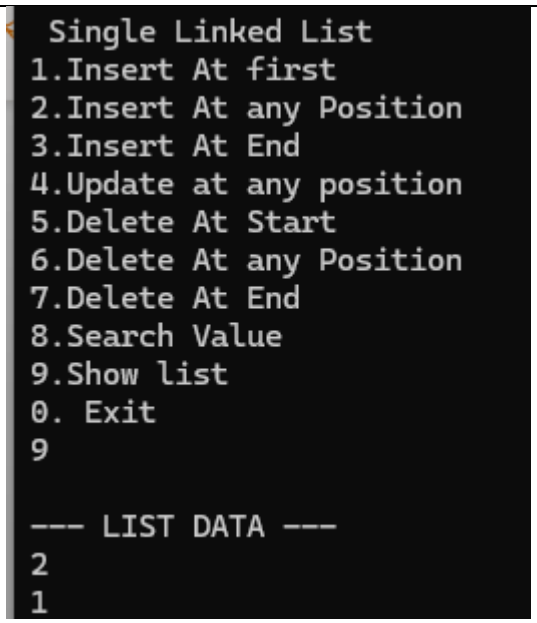
Enter Value you want to Search in the list : 1

1 Exist at Positon : 2
```

Task 5:	

Code:	void UpdateAtAnyPosition(string value,int pos)
	{
	node *tem = head;
	for(int i=0; i<pos-1; i++)
	{
	tem = tem->next;
	}
	tem->data = value;
	}

Screenshots :	--- LIST DATA --- 0 1 Single Linked List 1.Insert At first 2.Insert At any Position 3.Insert At End 4.Update at any position 5.Delete At Start 6.Delete At any Position 7.Delete At End 8.Search Value 9.Show list 0. Exit 4 Enter Value you want to update at certain position : 2 Enter Position you want to insert : 1



Complete Code:

```
#include <iostream>
using namespace std;

struct node{
 string data;
 node *next;
};

node *head = nullptr;

void insertAtStart(string value)
{
    node *temp =new node();
    temp->data = value;
    temp->next = head;
```

```
    head = temp;
}

void insertAtAnyPosition(string value,int pos)
{
    node *temp =new node();
    temp->data = value;

    if (pos == 1)
    {
      temp->next = head;
      head = temp;
      return;
    }

    node *tem = head;
    for(int i=0; i<pos-2; i++)
    {
       tem = tem->next;
    }
    temp->next = tem->next;
    tem->next = temp;
}

void insertAtEnd(string value)
{
    node *temp = new node();
```

```
    temp->data = value;
    temp->next = nullptr;

    if (head == nullptr)
    {
        head = temp;

    }else{

       node *tem = head;
        while(tem->next != nullptr)
        {
            tem = tem->next;
        }
        tem->next = temp;
    }
}

void UpdateAtAnyPosition(string value,int pos)
{
    node *tem = head;
    for(int i=0; i<pos-1; i++)
    {
       tem = tem->next;
    }
    tem->data = value;
 }
```

```
void DeleteAtStart()
{
   if(head == nullptr)
   {
      cout<<"List is Empty"<<endl;
      return;
   }
   else{
      node *Todelet = head;
      head = head->next;
      delete Todelet;
      return;
   }
}

void DeleteAtAnyPosition(int pos)
{
   if (pos == 1)
   {
     node *Delet = head;
    head = head->next;
    delete Delet;
    return;
   }

 else{
```

```
    node *prev = head;
    for(int i=0; i<pos-2; i++)
    {
      prev = prev->next;
    }
    node *Delet = prev->next;
    prev->next = Delet->next;
    delete Delet;
    return;
 }
}

void DeleteAtEnd()
{
    if(head==nullptr)
    {
          cout<<"List is Emty"<<endl;
          return;
    }

     if (head->next == nullptr)
    {
       delete head;
       head=nullptr;
       return;

    }else{
```

```
    node *prev = head;
    node *curr = head->next;
     while(curr->next != nullptr){
        prev = curr;
        curr = curr->next;
     }
         prev->next = nullptr;
        delete curr;
  }
}

void SearchValue(string value)
{
   node *temp = head;
   int pos = 1;

  while(temp!=nullptr)
  {
     if(temp->data == value)
   {
     cout<<value<< " Exist at Positon : "<< pos<<endl;
     return;
   }
   temp = temp->next;
   pos++;
  }
```

```
   cout<<value<<" does not exist. "<<endl;
}

void showlist()
{
    node *temp = head;
    if (temp == nullptr)
    {
        cout << "List is empty!" << endl;
        return;
    }
    while(temp != nullptr)
    {
        cout<<(temp->data)<<endl;
        temp = temp->next;
    }
}

int main()
{
    int x,pos;
    string value;

    do{
    cout << " Single Linked List " << endl;
    cout << "1.Insert At first" <<endl;
    cout << "2.Insert At any Position " <<endl;
```

```
cout << "3.Insert At End "<<endl;
cout << "4.Update at any position "<<endl;
cout << "5.Delete At Start "<<endl;
cout << "6.Delete At any Position "<<endl;
cout << "7.Delete At End "<<endl;
cout << "8.Search Value"<<endl;
cout << "9.Show list " <<endl;
cout << "0. Exit "<<endl;
cin >> x;
cout<<endl;

if(x<0 || x>9)
{
    cout<< "Invalid input! TRY AGAIN "<<endl;
}
if (x==1)
{
   cout<<"Enter Value you want to insert at first : ";
   cin>>value;
   cout<<endl;
   insertAtStart(value);
   cout<<endl;
}
 if (x==2)
{
   cout<<"Enter Value you want to insert at certain position : ";
   cin>>value;
```

```
    cout<<endl;
    cout<<"Enter Position you want to insert : ";
    cin>>pos;
    insertAtAnyPosition(value,pos);
    cout<<endl;
  }

  if (x==3)
  {
    cout<<"Enter Value you want to insert at end : ";
    cin>>value;
    cout<<endl;
    insertAtEnd(value);
    cout<<endl;
  }

  if (x==4)
  {
    cout<<"Enter Value you want to update at certain position : ";
    cin>>value;
    cout<<endl;
    cout<<"Enter Position you want to insert : ";
    cin>>pos;
    UpdateAtAnyPosition(value,pos);
    cout<<endl;
  }
```

```
if (x==5)
{
  cout<<"The first value is deleted : "<<endl;
  DeleteAtStart();
  cout<<endl;
}

if (x==6)
{
  cout<<"Enter Position which you want to delete : ";
  cin>>pos;
  DeleteAtAnyPosition(pos);
  cout<<endl;
}

  if (x==7)
{
  cout<<"The last value is deleted : ";
  DeleteAtEnd();
  cout<<endl;
}

 if (x==8)
{
cout<<"Enter Value you want to Search in the list : ";
  cin>>value;
  cout<<endl;
```

```
      SearchValue(value);
      cout<<endl;
    }

    if (x==9)
    {
       cout<<"--- LIST DATA ---"<<endl;
       showlist();
    }
    }
    while(x!=0);

    cout<<"Program Ends!Thank you " <<endl;

    return 0;
}
```